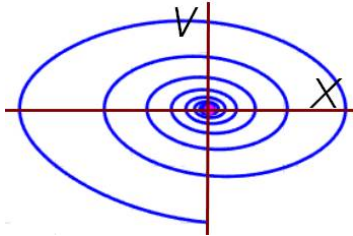


- 7 (a) Max v at $x = -58$ m where grad is max [1]
 Gradient = $-78/(6.8-3.2) = -21.7 \text{ m s}^{-1}$ [1]
(Accept range 18-26 m s^{-1})

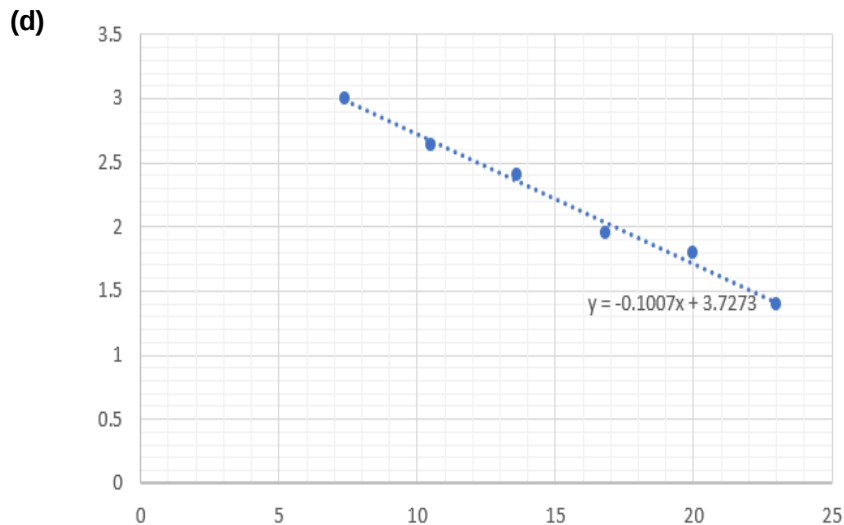
- (b)  Clockwise spiral, starting from $-V_{\text{max}}$ and $X = 0$ [1]
 Decreasing max speed and amplitude [1]
Note: Give 1 m for anticlockwise spiral

- (c) (i) Position where net force/net acceleration = 0 [1]
DO NOT ACCEPT: Upward tension by cord = downward weight of cuboid.

(ii)

Time / s	Position / m	Magnitude of displacement, A / m	$\ln(A/m)$
7.4	- 78	20	3.00
10.5	- 44	14	2.64
13.6	- 69	11	2.40
16.8	- 51	7	1.9
20.0	- 64	6	1.8
23.0	- 54	4	1.4

Amplitude column [1]
 $\ln(A/m)$ column [1]
(ignore sf totally)



- Plot $\ln A$ vs t [1]
 $-c = \text{gradient} = \text{working with coordinates}$ [1] (range 0.09 – 0.11)
 $c = + 0.1007 \text{ s}^{-1}$ [1]
 $A_0 = \exp(\text{intercept})$ or substitution into equation. [1]
 $A_0 = 41.7 \text{ m}$ [1] (range 35 – 45)

(e) cuboid is not oscillating / cuboid is free falling [1]

(f) (i) From the graph, $T = 6.2 \text{ s}$ (accept $6.0 - 6.4$) [1]

(ii) Using $\frac{4\pi^2}{T^2} = \frac{k}{m} - (1.00 \times 10^{-2})$, $k = 63.46 \text{ N m}^{-1}$ [1]

Hence, at equilibrium, $mg = ke$

Solving, we get $e = 9.46 \text{ m}$ [1]

Note:

$T = 6.0$, get $k = 67.68$ and $e = ???$

$T = 6.1$, get $k = ???$ and $e = ???$

$T = 6.3$, get $k = 61.44765$ and $e = 9.76$

$T = 6.4$, get $k = 59.561$ and $e = 10.1$

(iii) Gain in EPE = $\frac{1}{2} ke^2 = 2838 \text{ J}$
Loss in GPE = $mgh = -34800 \text{ J}$
Gain in KE = $\frac{1}{2} (600/9.81)(21.7)^2 = 14400 \text{ J}$

[Give 2 marks for any 2 correct energy changes]

Change in total energy = $-34800 + 14400 + 2838$
= -17562 J
= -17600 J [1]

(g) Thicker cord increases the elastic constant and hence prevents the heavier mass from over-extending the cord, improving safety. (accept cord may snap) [1]